Hsin Jee Sum

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EDUCATION

University of Leeds **Leeds, UK**
MEng, BSc Computer Science with Artificial Intelligence Sep 2022 – Jul 2026

- Average Grade: 75% (First Class Honours)

WORK EXPERIENCE

Leeds Gryphon Racing, Formula Student **Leeds, UK**
Visions Team Lead; AI R&D Division Sep 2025 – Jun 2026

- Led 10+ developers across sprints to deliver autonomy stack, placing 24th of 59 teams at FS-UK 2026 (193.5 points)
- Reduced inference 10% to sub-20ms at 92% mAP on an inherited YOLOv8 vision model for live object detection
- Cut on track testing by 30% via Gazebo simulations with 40+ C++ unit tests for perception modules at 80% coverage

SRKK Group **Kuala Lumpur, MY**
Summer Intern; Software Development (RAG) Aug 2025 – Sep 2025

- Built an Azure Power Automate RAG pipeline for 1000+ page financial RFPs, cutting human review time by 70%
- Lowered LLM hallucination by 30% through custom faithfulness scoring, grounding answers in retrieved chunks
- Chunked documents using recursive character splitting, cutting query latency to 200ms and cutting token use by 40%

EH Integrated Systems **Kuala Lumpur, MY**
Summer Intern; Full-Stack Software Engineer Jul 2024 – Sep 2024

- Cached invoice reads to hold checkout, refund, and invoicing under 1% errors and ~100ms at 200K queries/day
- Containerized C# services with Docker preventing duplicate setup, cutting environment time by ~85%
- Designed a 12-table MySQL schema with indexes across 1M+ records, cutting inventory query latency to ~20ms

PROJECTS

Automated methods for AI Video Detection *(Master's Thesis Project)* Sep 2025 – May 2026

- Achieved 97% precision, 0.9 AUC building a Video Swin Transformer for synthetic video detection (team of 5)
- Cut training epoch time by 35% on the shared AIRE HPC by staging data to flash, with I/O wait down 40%
- Led comparative analysis of 8 unique architectures; final ensemble model achieved highest recall (0.62)

CNN for Breast Cancer Classification *(Bachelor's Thesis Project)* Sep 2024 – May 2025

- Derived and coded backprop, ReLU activations, and weight updates by hand for a breast tumor CNN in NumPy
- Accelerated convergence time by 30% by designing custom Batch Normalization layers, stabilizing gradient flow
- Mitigated class imbalance using weighted loss functions and 5-fold CV, achieving 90.65% recall and 88.7% F1 score

Hand-Sign Computer Vision & 3D Animation Engine Feb 2026

- Built a TensorFlow.js model (99% acc.) and a Three.js 3D engine rendering each sign as 500K+ swarming particles

EXTRACURRICULAR ACTIVITIES

Open-source contributor: Dify (langgenius/dify) Oct 2026 – Current

- Submitted 3 open PRs to Dify replacing mocks in 4 tests with real DB fixtures; added 1 tenant-isolation test
- Broke covered logic on purpose in all 3 PRs to prove tests fail; 209 surrounding tests and ruff pass

DurHack Hackathon 2025 | Mario Kart Win Predictor **Durham, UK**
G Research Prediction Challenge, Honorable Mention Nov 2025

- Predicted game race outcomes at 99% accuracy from 5000+ character, kart, map, and tire stats via TensorFlow model
- Achieved Honorable Mention by G Research among over 100 teams (top 2%) as part of a 3-person team

British Chinese Society Treasurer **Leeds, UK**
Leeds University Society Sep 2023 – Jun 2024

- Managed a £2,800 budget for 200 members, allocating funds to raise £600 in sponsorship and hold 6 events on plan

ADDITIONAL INFORMATION

Languages:	English (Native), Mandarin (Native), Cantonese (Native), Malay (Intermediate), Japanese (Basic)
Technical Skills:	C#, Python, Java, C, C++, JavaScript, SQL, Go, TensorFlow, PyTorch, Scikit-learn, Docker, Azure
Interests:	Grade 7 Piano, Grade 6 Violin, Leeds Badminton Club Team 2, Open-source Contributing